

Case Study: Andy Reid and the Kansas City Chiefs

A West Coast coach, a college playbook, a quarterback who broke the clock, and a defensive coordinator who knew when to send everyone.

Super Bowl LIV, February 2, 2020, in Miami Gardens. Kansas City trails San Francisco 20–10 with 7:13 left, and it is third-and-15 at the Chiefs’ own 35. The 49ers have the best pass rush in the building and a coverage that keeps everything in front of it. A completion short of 15 yards is worth nothing, and an incompletion means a punt with seven minutes left and the game slipping away. On the sideline Patrick Mahomes asks his coaches whether there is time to run a play called “Wasp.” There is. He takes the snap, retreats, keeps retreating, and from **14 yards behind the line**, with a 49ers defensive tackle arriving, throws the ball 57 yards through the air to Tyreek Hill near the left sideline. Forty-four yards. Kansas City scores three touchdowns in the last seven minutes and wins its first Super Bowl in fifty years.¹ Why did Andy Reid’s offense have a play built for third-and-15? Why would a quarterback ask for the slowest-developing route on the call sheet, with the league’s best rush coming? And how did a 61-year-old coach who learned his football in Green Bay in the early 1990s end up running the most modern offense in the sport? This chapter is about the team that answered those questions better than anyone for seven years, and about what happened when opponents found answers of their own.

You’ll be able to...

- Explain how Andy Reid built the Chiefs’ offense from three sources: West Coast bones, college spread and RPO ideas, and a playbook full of plays invented for one opponent, and spot each on a broadcast.
- Explain, with numbers, why a quarterback who stays dangerous after four seconds changes the defense’s arithmetic, and what defenses do about it.
- Contrast the Tyreek Hill era (2018–2021) with the Travis Kelce quick-game era (2022–2024), and read the option route against man and against zone.
- Explain Steve Spagnuolo’s pressure defense as the complement to the offense, and recognise his Cover 0 before the ball is snapped.
- Name the offensive coordinators by season (Eric Bieniemy 2018–2022, Matt Nagy 2023–2025, Bieniemy again in 2026) and the five Super Bowls (won LIV, LVII and LVIII; lost LV and LIX), with what decided each.
- Explain how opponents answered (two-high shells, three-man rushes, four-man pressure with no blitz) and what survives of the Chiefs’ approach after the 6–11 season of 2025.

¹“Jet Chip Wasp,” Wikipedia: https://en.wikipedia.org/wiki/Jet_Chip_Wasp (the full call “2-3 Jet Chip Wasp”; Mahomes asking whether there was time to run “Wasp”; Hill’s route “leading defenders towards the middle of the field before cutting sharply towards the sideline”; Mahomes 14 yards behind the line; final score 31–20). CBS Sports, “‘Third-and-forever’: How Patrick Mahomes, Tyreek Hill ignited Chiefs’ Super Bowl 2020 comeback with one play” (Feb. 2020; third-and-15 from the Kansas City 35, trailing 20–10 with 7:13 left; 57.1 yards in the air for a 44-yard gain). Kansas City’s previous Super Bowl title was Super Bowl IV, after the 1969 season. URL: <https://www.cbssports.com/nfl/news/third-and-forever-how-patrick-mahomes-tyreek-hill-ignited-chiefs-super-bowl-2020-comeback-with-one-play> . Super Bowl IV was played January 11, 1970, so LIV (February 2, 2020) came fifty years later. Reid’s age (61 at Super Bowl LIV) from his March 19, 1958 birth date, Wikipedia (cited in the note on his career).

i You'll need

This is a Level 4 case study, so it leans on a lot of earlier chapters. One line each:

- **The modern era** (The Modern Era): after 2017 the league's offenses went to motion, condensed sets and play-action, and defenses answered by playing more two-high coverage (two deep safeties) to take away big plays. The Chiefs are one of that chapter's main characters; here they get the full treatment.
- **Blitzing** (Blitzing: Numbers, Overloads, A-Gap Mugs, and Cover 0): a blitz sends more rushers than the protection can block, hoping an unblocked rusher arrives before the ball leaves. A Cover 0 blitz sends six or more and plays Cover 0 behind it: man-to-man on every receiver, nobody deep. A double A-gap mug puts two linebackers in the gaps either side of the center to make the protection guess.
- **Systems and play calls** (Offensive Systems and Play-Calling Languages): the West Coast offense (WCO) is Bill Walsh's family of short, timed, horizontal throws and long, descriptive play calls; the Air Raid is the college pass offense that taught a generation of quarterbacks, Mahomes included.
- **RPOs and the quick game** (Run-Pass Options, The Quick Game): an RPO is a run play with a pass attached, decided by the quarterback's read of one defender; the quick game is throws released in about two seconds or less.
- **Routes** (Passing Fundamentals): an option route lets the receiver choose his break from what the coverage does; air yards are how far past the line a pass travels in the air; a chip is a back or tight end hitting a pass rusher before releasing into his route (Pass Protection).
- **Off-script football** (Quarterback Play): an off-script play is one that leaves its design, usually because the quarterback leaves the pocket; the scramble drill is the set of rules receivers follow when he does.
- **Coverage** (Man Coverage, Cover 3): in man coverage each defender has a receiver; Cover 1 is man with one deep safety; in Cover 3 three defenders each take a deep third and four play zones underneath. Leverage is which side of a receiver a defender stands on.
- **The numbers** (Expected Points and Success Rate): EPA (expected points added) is how much a play changed the offense's expected score; average NFL plays sit close to zero, and a team that averages +0.15 a play over a season is elite.

Reading the diagrams: offense is blue circles, defense outlined orange squares. Where a real play is drawn, Chiefs players wear their jersey numbers (15 Mahomes, 10 Hill, 87 Kelce); otherwise the offense uses letters (Q quarterback, R running back, X, Z and H receivers, Y tight end) and the defense uses E and T for linemen, M and W for linebackers, \$ for the nickel back, CB for cornerbacks and FS and SS for safeties. A green ring marks the player to watch. The blue line is the line of scrimmage, the orange line the line to gain, and the faint lines across the field are 5 yards apart.

The problem: a West Coast coach in a league that was changing

Andy Reid was born in Los Angeles in 1958, played offensive tackle at Brigham Young under LaVell Edwards (whose teams threw the ball when almost nobody in college did), and coached offensive linemen at four colleges from 1983 to 1991 before Mike Holmgren hired him in Green Bay in 1992. Holmgren was one of Bill Walsh's assistants from the 49ers, so Reid learned the West Coast offense from its second generation: tight ends coach and assistant line coach first, then quarterbacks coach for Brett Favre in 1997 and 1998. In January 1999 the Philadelphia Eagles made him, at 40, the second-youngest head coach in the league. He stayed fourteen seasons, won 130 regular-season games, reached five NFC Championship Games (four in a row from 2001 to 2004) and lost Super Bowl XXXIX to New England, 24–21. After a 4–12 season the Eagles let him go on the last day of 2012. Four days later Kansas City hired him.²

The team he took over had gone 2–14. The first five Kansas City seasons were the kind most coaches would frame: 11–5, 9–7, 11–5, 12–4 and 10–6, with Alex Smith, a quarterback acquired from San Francisco, playing efficient, careful football. They also carried a ceiling you can see in the playoff results. A 45–44 loss at Indianapolis in January 2014, after leading by 28 in the third quarter. A 27–20 loss at New England two years later. An 18–16 home loss to Pittsburgh. A 22–21 home loss to Tennessee in January 2018,

²“Andy Reid,” Wikipedia: https://en.wikipedia.org/wiki/Andy_Reid (born March 19, 1958; Glendale Community College 1976–1977 and BYU 1978–1980 as an offensive tackle; BYU graduate assistant 1982; offensive line coach at San Francisco State 1983–1985, Northern Arizona 1986, UTEP 1987–1988 and Missouri 1989–1991; Packers assistant offensive line and tight ends coach from 1992 and quarterbacks coach and assistant head coach 1997–1998; hired by the Eagles January 11, 1999, “the second-youngest head coach in the league” after Jon Gruden; 130–93–1 regular season and 10–9 in the playoffs with Philadelphia; five NFC Championship Games, four in a row 2001–2004 plus 2008; Super Bowl XXXIX lost 24–21 to New England; 4–12 in 2012, his worst Eagles record; Jeffrey Lurie announced on December 31, 2012 that his contract would not be renewed; five-year agreement with Kansas City on January 4, 2013; in 2017, after the Chiefs lost six of seven, Reid “conceding play calling duties to offensive coordinator Matt Nagy”). Holmgren was a 49ers assistant under Walsh before taking the Green Bay job: Case Study: Walsh's 49ers. Pederson's Super Bowl LII title (after the 2017 season) and Nagy's hiring by Chicago in January 2018 are covered in the note on Reid's coaching tree.

after leading 21–3 at halftime.³ In those years the Chiefs’ offense ranked between 9th and 15th in the league in EPA per play: good, never frightening (the chart in the data-fingerprint section below shows the whole run).

That ceiling had a schematic shape. The classic West Coast offense is a machine for completing passes: three- and five-step drops, routes timed to the quarterback’s last step, throws spread across the width of the field so that a 6-yard completion works like a run. It needs a quarterback who throws on time and accurately, and Smith did. What it doesn’t do by itself is produce explosive plays, and by the middle 2010s NFL defenses had grown very good at the thing a timing offense needs least: keeping everything in front of them and tackling the catch. A team that has to drive 75 yards in twelve completed passes gives a defense twelve chances to win a down. Reid’s problem in Kansas City was how to keep the efficiency and add the explosions, without blowing up the system his staff and players already knew.

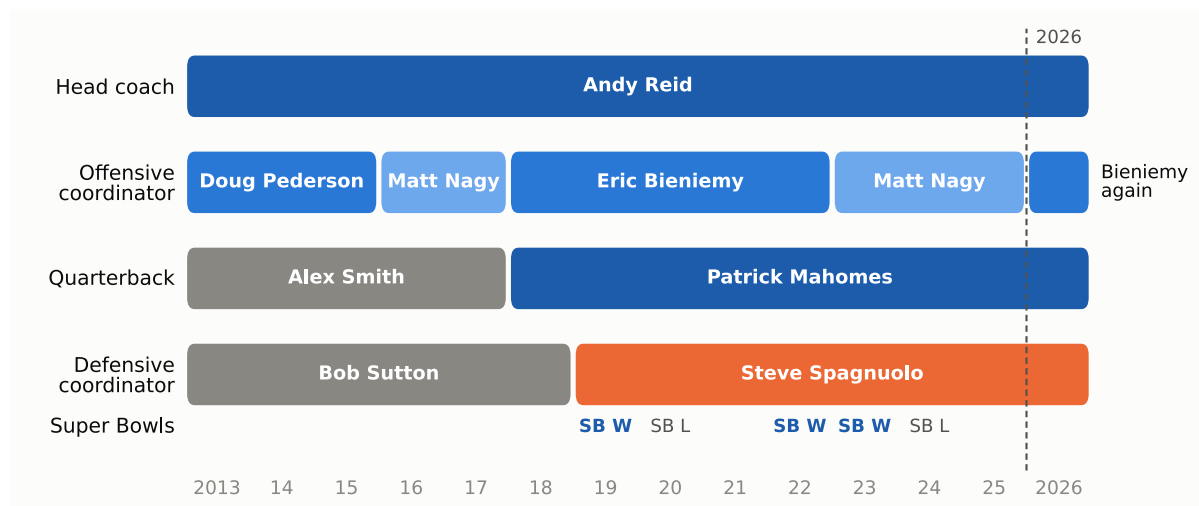


Figure 1: The Reid years in Kansas City at a glance. Reid has been head coach since 2013; he calls the plays himself (except for a stretch of 2017 when he handed them to Matt Nagy), and his offensive coordinator relays and helps build them. Coordinators: Doug Pederson (2013–2015), Matt Nagy (co-coordinator with Brad Childress in 2016, sole coordinator in 2017), Eric Bieniemy (2018–2022), Nagy again (2023–2025) and, from 2026, Bieniemy again. Defense: Bob Sutton (2013–2018), Steve Spagnuolo (2019 to now). The Mahomes years produced five Super Bowl trips in six seasons (W = won, L = lost).

A word on who does what, because it confuses almost everyone. Reid is the play-caller: he chooses the play and says a short name for it into his headset. The offensive coordinator finds it on his own copy of the sheet and relays the full call to the quarterback (The Game Plan describes the two-step relay and Reid’s famous oversized play card). The coordinator also builds the weekly plan, coaches the quarterbacks and runs meetings, which is why Reid’s coordinators keep leaving for head-coaching jobs: Doug Pederson won a Super Bowl as Philadelphia’s head coach two seasons after leaving, and Matt Nagy became the Bears’ head coach after 2017. The one exception to Reid calling plays came in late 2017, when a slump of six losses in seven games led him to hand the calls to Nagy for the rest of the season.

The scheme answer, part 1: keep the West Coast bones

Reid didn’t throw the old offense away. Watch a Chiefs game from any season and you will still see its skeleton:

- **Horizontal stretches.** Routes that spread defenders sideways, so a linebacker can’t cover both a flat route and a short in-cut. The quick game (The Quick Game) is pure West Coast, and so are the running back’s checkdowns.
- **The tight end and the back as receivers.** Walsh’s offenses threw to everybody. Reid’s have always fed a tight end (Brent Celek and the young Zach Ertz in Philadelphia, Travis Kelce in Kansas City) and a receiving back.
- **Long, descriptive calls.** “2-3 Jet Chip Wasp” is a West Coast sentence: protection, a word for the back, a route name. The language is the family resemblance, even when the plays look nothing like Walsh’s (Offensive Systems shows the call broken into its parts).

³Course calculation from nflverse schedules ([load_schedules](#)), regular seasons and playoffs, 2012–2025: Kansas City went 2–14 in 2012; 11–5 (2013), 9–7 (2014), 11–5 (2015), 12–4 (2016) and 10–6 (2017). Playoffs: lost 45–44 at Indianapolis (Jan. 4, 2014); beat Houston 30–0, then lost 27–20 at New England (January 2016); lost 18–16 to Pittsburgh (Jan. 15, 2017); lost 22–21 to Tennessee (Jan. 6, 2018). The 28-point lead in Indianapolis (38–10, 76 seconds into the third quarter): Associated Press, “Colts erase 28-point deficit” (Jan. 4, 2014), via *The Washington Times*: <https://www.washingtontimes.com/news/2014/jan/4/nfl-playoffs-colts-erase-28-point-deficit-and-elim/>, and “2013 Kansas City Chiefs season,” Wikipedia (“after blowing a 38–10 lead”). The 21–3 halftime lead against Tennessee: “2017 Kansas City Chiefs season,” Wikipedia (“a commanding 21–3 lead at halftime”). Offensive EPA ranks: see the note on the EPA chart.

- **A big menu, installed weekly.** Reid draws new plays on index cards early in the week, and his opening script is built from them (The Game Plan tells the story). That habit comes from Walsh too, who scripted his openers and treated the playbook as something rebuilt for each opponent (Case Study: Walsh's 49ers).

! Why it exists

The West Coast core is what makes the rest possible. A team that can complete a 6-yard pass on any down doesn't fall behind schedule, and a team that stays on schedule can afford to call its odd, risky, creative plays when *it* chooses, not when it is desperate. The bones are the insurance policy. What they give up is speed to the end zone: a pure timing offense needs many plays to score, and every play is another chance for a sack, a penalty or a turnover.

The scheme answer, part 2: graft the college game onto it

The second source arrived on purpose. When Reid came to Kansas City in 2013 he brought Brad Childress, his old Philadelphia coordinator and former Vikings head coach, and gave him a title the NFL had never seen: *spread game analyst*. Childress's job for three seasons was to study the college spread offenses of the time (the zone read, the packaged plays that would soon be called RPOs, the empty and trips formations) and work out what would survive in the NFL. In 2016 Childress and quarterbacks coach Matt Nagy became co-coordinators; in 2017 Nagy had the job alone, and the Chiefs built a large part of their offense on run-pass options, years before most of the league did.⁴

You know the RPO from Run-Pass Options: the linemen block a run, and the quarterback, holding the ball against the running back's belly at the mesh (the instant of the possible handoff), watches one defender who can't do two jobs. If that defender comes down to stop the run, the quarterback pulls the ball and throws into the space he just left. If he stays out to cover the pass, the back gets the ball against a box the offense has blocked.

⁴NFL.com, "Brad Childress joins Chiefs as spread game analyst" (March 2013): <https://www.nfl.com/news/brad-childress-joins-chiefs-as-spread-game-analyst-0ap1000000155148> ; "Brad Childress," Wikipedia: https://en.wikipedia.org/wiki/Brad_Childress (spread game analyst 2013–2015, offensive coordinator 2016, assistant head coach 2017); "Matt Nagy," Wikipedia: https://en.wikipedia.org/wiki/Matt_Nagy (co-offensive coordinator with Childress in 2016, offensive coordinator in 2017; Bears head coach 2018–2021). On the 2017 RPO volume ("they ran 24" against Oakland, counting one play flagged for an ineligible man downfield): Arrowhead Pride, "Chiefs OC Matt Nagy is making minor changes for huge results" (Dec. 14, 2017): <https://www.arrowheadpride.com/2017/12/14/16775348/chiefs-oc-matt-nagy-is-making-minor-changes-for-huge-results> . Smith's 2017 league-leading passer rating (104.7) and Hunt's rushing title (1,327 yards, the sixth rookie of the Super Bowl era to lead the league): Matt McMullen, Chiefs.com, "Several Chiefs Etched Their Names in the Record Books in 2017" (Jan. 17, 2018): <https://www.chiefs.com/news/several-chiefs-etched-their-names-in-the-record-books-in-2017-20266838> . Reid handing play-calling to Nagy in 2017: "2017 Kansas City Chiefs season," Wikipedia. "Years before most of the league" is a characterisation: no public charting counts RPOs before FTN's begins in 2022. Offensive EPA ranks: course calculation (see the EPA chart).

the shotgun on 81% of their snaps that season, against a league average of 64%, and stayed around 80% for the next seven years: about as far from the under-center Shanahan offenses of the same era as a West Coast descendant could get (Case Study: The Shanahan Tree).

The scheme answer, part 3: the menu, and the plays nobody else has

The third source is Reid himself. On the first-quarter drive of Super Bowl LIV that put Kansas City ahead, facing fourth-and-1 at the San Francisco 5, the Chiefs lined up with three players stacked behind Mahomes; Mahomes turned a full circle, the others spun and reset, and running back Damien Williams took a direct snap and ran for 4 yards and a first down. Reid had found the play on old film of the 1948 Rose Bowl, Michigan against USC, and the Chiefs had been practicing it since spring.⁷ Three years later, two fourth-quarter touchdowns in Super Bowl LVII came on a return-motion play called “Corn Dog,” and a year after that the winning touchdown of Super Bowl LVIII came on a relative of it built for a defense that had studied Corn Dog (Motion and Shifts and What Beats What tell those two stories).

Common misconception

“The Chiefs win with trick plays.” They win with an ordinary core and use the strange plays as seasoning. In FTN’s charting, a trick play (reverses, flea-flickers, throwback passes and the like) was 1.5% of Kansas City’s plays in 2022, fourth in the league, but 0.5% in 2023 (26th), 0.9% in 2024 (9th) and 0.1% in 2025 (30th).⁸ What the Chiefs did far more than almost anyone was throw screens: between 6% and 10% of their plays from 2022 to 2025, never lower than seventh in the league. Screens are the West Coast’s way of punishing a pass rush, and they are the trick plays that work every week. The famous plays are memorable because they are rare; the screens are invisible because they are constant.

What the big menu buys is subtler than surprise. A defense prepares for an opponent by studying its tendencies: from this formation, with this motion, they run this. Reid’s offense shows the same handful of concepts from so many formations, shifts and motions that the tendency table gets thin. The illusion of complexity (Constraint Theory and Sequencing) usually describes a small playbook dressed up to look big. Reid runs a genuinely big one, and his players carry it because the core underneath doesn’t change.

Patrick Mahomes and the defensive arithmetic

Every pass defense is built on a clock. A four-man rush reaches an NFL quarterback, on average, in something under three seconds; the coverage has to hold for that long, and then the rush finishes the play. Defensive coordinators design coverages around that budget: zone defenders drop to spots they can hold for two and a half or three seconds, man defenders play technique they can sustain for that long. When the clock runs out, the quarterback has to throw it away, take a sack or improvise, and for almost everyone improvising is a losing bet.

Here is that bet in numbers. From 2018 to 2025, NFL passes thrown in under four seconds averaged about +0.17 EPA per attempt. Passes thrown after four seconds or more averaged about −0.03: a bad play, on average, before you even count the sacks the long-held ball produces. Now look at one quarterback.

⁷Adam Teicher, ESPN, “1948 Rose Bowl? Chiefs coach Andy Reid pulls a play from 72 years ago” (Feb. 4, 2020; Reid saw the play on film of Michigan’s 49–0 win over USC): https://www.espn.com/blog/kansas-city-chiefs/post/_id/27840/1948-rose-bowl-chiefs-coach-andy-reid-pulls-a-play-from-72-years-ago ; NFL.com, “Chiefs’ spin play in SB LIV lifted from 1948 Rose Bowl”: <https://www.nfl.com/news/chiefs-spin-play-in-sb-liv-lifted-from-1948-rose-bowl-0ap3000001101036> (practiced since the spring; Mahomes: “We’ve worked on that play since the first day of OTAs Phase II”). Play-by-play: nflverse, game 2019_21_SF_KC, first quarter, 1:57, “Direct snap to 26-Dam.Williams ... up the middle to SF 1 for 4 yards” on fourth-and-1 from the San Francisco 5.

⁸Course calculation from FTN charting (`load_ftn_charting`) joined to nflverse play-by-play, regular-season pass and run plays. `is_trick_play`: Kansas City 1.48% in 2022 (4th of 32), 0.47% in 2023 (26th), 0.94% in 2024 (9th), 0.10% in 2025 (30th). `is_screen_pass`: 7.4% (2nd), 9.8% (1st), 7.6% (4th), 6.0% (7th). `is_rpo`: 11.4% (2nd), 6.4% (6th), 6.1% (3rd), 14.4% (1st), against league medians of 3.4% to 5.7%. FTN’s definitions are its own charters’ judgments.

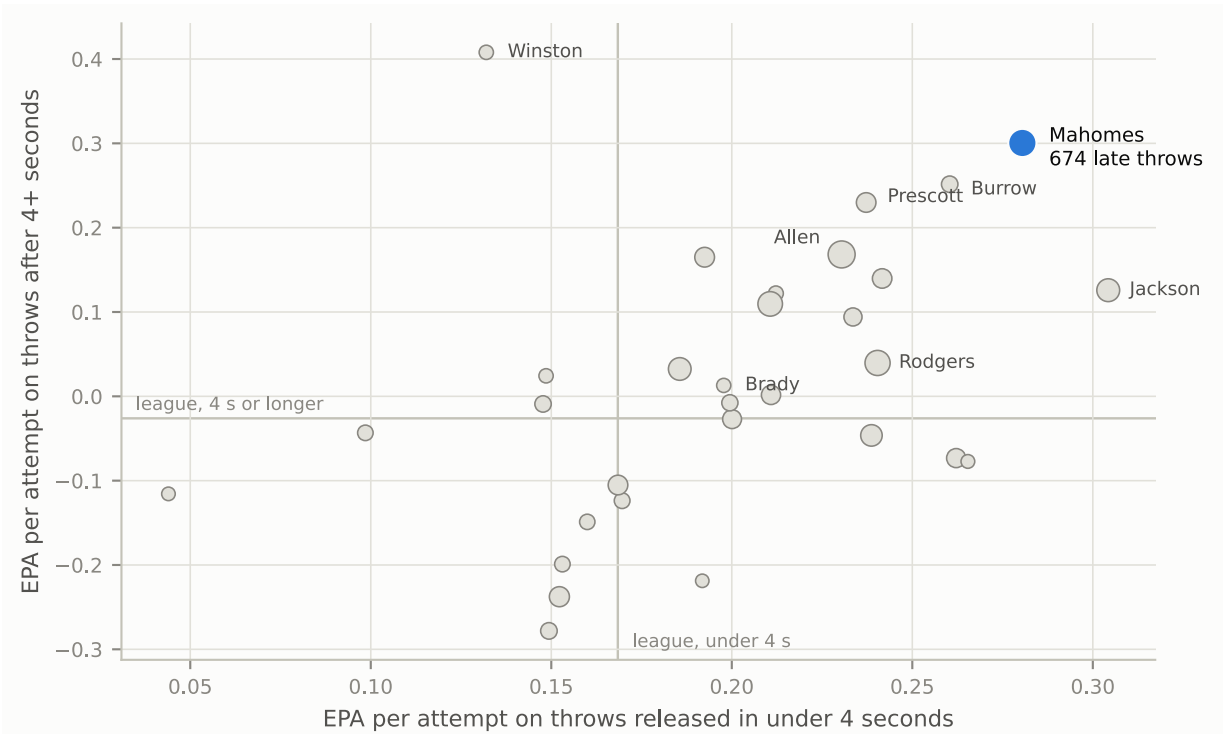


Figure 3: Throwing late. Every quarterback with at least 150 pass attempts released four or more seconds after the snap, 2018–2025 regular seasons (dots sized by those late throws). Horizontal: EPA per attempt on throws released in under four seconds; vertical: on throws released after four seconds or more. The grey lines are the league averages: about +0.17 for quick throws and –0.03 for late ones, so for the league as a whole the long-held ball is a losing play. Mahomes is in the top-right corner: his late throws (674 of them, about 15% of his attempts against a league-wide 10%) were worth as much as his on-time throws, and as much as most quarterbacks’ best. Jameis Winston’s late throws rate higher on a much smaller, mostly 2018–2019 sample. Sacks are not included.

Data: nflverse play-by-play and NGS time to throw (2018–2022) and FTN (2023–2025) via nflreadpy.

Patrick Mahomes’s throws after four seconds, all 674 of them from 2018 to 2025, were worth about +0.30 EPA each: slightly *more* than his on-time throws (+0.28), and about a third of a point more per throw than the league’s late throws.⁹ He threw late about half again as often as the average quarterback, and it didn’t cost him. FTN’s charting from 2022 says the same thing about leaving the pocket: Mahomes was outside it on about 23% of his dropbacks through 2025 (the league, 18%), and those dropbacks were worth +0.31 EPA each, more than double his in-pocket plays, while the league’s out-of-pocket dropbacks were worth roughly nothing.¹⁰ (Quarterback Play has the league-wide picture and the left-handed flip in Denver that Reid calls Mahomes’s best play.)

Think about what that does to a defensive coordinator’s arithmetic. Every coverage call is a bet that the rush gets home before the coverage breaks. Against most quarterbacks the bet pays twice: the quick throws are contested, and the late ones are bad. Against Mahomes, the late half of the bet doesn’t pay. So the defense has to change something, and every change costs:

- **Cover longer.** Play coverage that holds for five seconds, not three. That usually means more defenders in coverage and fewer rushing, which gives the quarterback even more time.
- **Rush with discipline.** Keep the rushers in rush lanes so he can’t escape outside, instead of winning with speed around the edge. That gives up the fastest route to a sack.
- **Spy him.** Assign one defender to mirror him (spy). That defender is not covering anyone.
- **Get there sooner.** Blitz, so the ball has to come out before four seconds. That leaves fewer defenders in coverage, and against a quarterback who sees the field, more open receivers.

⁹Course calculation from nflverse play-by-play joined to `load_participation()` `time_to_throw` (NGS through 2022, FTN from 2023), 2018–2025 regular seasons, pass attempts (sacks and spikes excluded): league +0.168 EPA per attempt on throws under 4.0 seconds (129,293 attempts) and –0.026 on throws of 4.0 seconds or more (13,810); Mahomes +0.280 (3,948) and +0.300 (674), 14.6% of his attempts late against 9.7% league-wide. Among the 31 quarterbacks with 150 or more late throws he ranks second, behind Jameis Winston (+0.408 on 172). Pass-rush timing (“under three seconds”): the league’s median time to throw in 2025 was about 2.5 seconds; see Disguise, Rotation, and Split-Field Coverage.

¹⁰Course calculation from FTN charting (`is_qb_out_of_pocket`) joined to nflverse play-by-play, 2022–2025 regular-season dropbacks (scrambles and sacks included): Mahomes left the pocket on 22.7% of dropbacks (league 18.0%) and averaged +0.305 EPA per dropback outside it and +0.144 inside; the league averaged +0.005 outside and +0.032 inside.

None of these is free, and the Chiefs' offense was built to collect on each one. The four-second plays aren't luck. They come from the scramble drill: rules every receiver follows the moment the quarterback leaves the pocket, so the improvisation is organised on both ends.

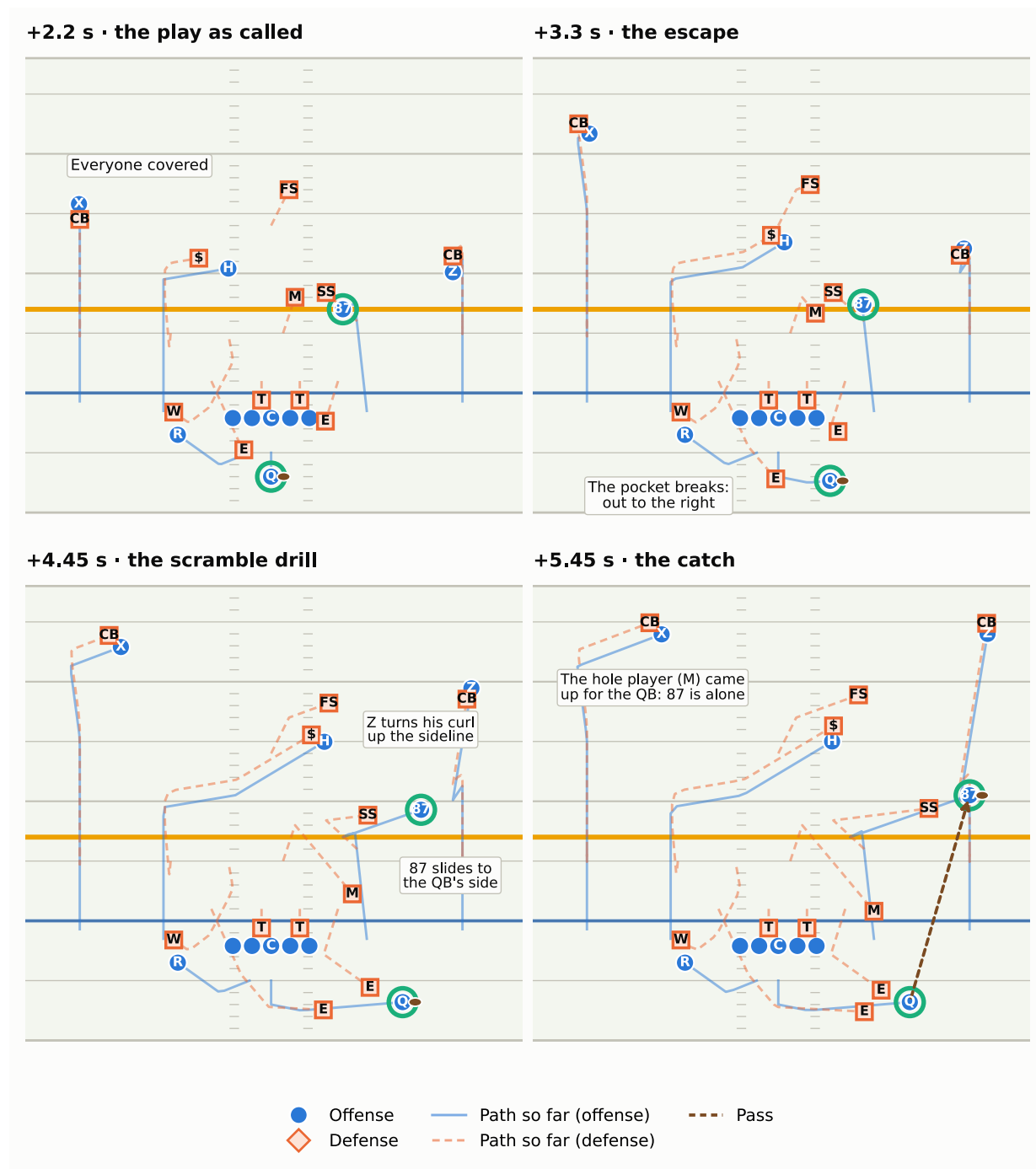


Figure 4: The scramble drill (an illustrative play, not a specific one): four rushers and man coverage (Cover 1), with the free safety deep and the Mike linebacker (M) sitting in the hole underneath. At 2.2 seconds every receiver is covered and the ball should be gone or thrown away. Instead the left end (E) beats the tackle, the quarterback slides away from him and breaks out to his right. The receivers follow the drill's rules: the deep receivers keep climbing (X drifts toward the quarterback's side), the intermediate receivers work back toward the sideline he is running to (Z turns his curl up the sideline; the tight end, 87, slides out along with him), and the crossing route (H) keeps crossing. The defenders, playing man with their backs to the quarterback, have to turn and chase, and the Mike has to choose between his zone and the quarterback. He comes up, and 87 is alone near the sideline when the ball arrives.

! Why it exists

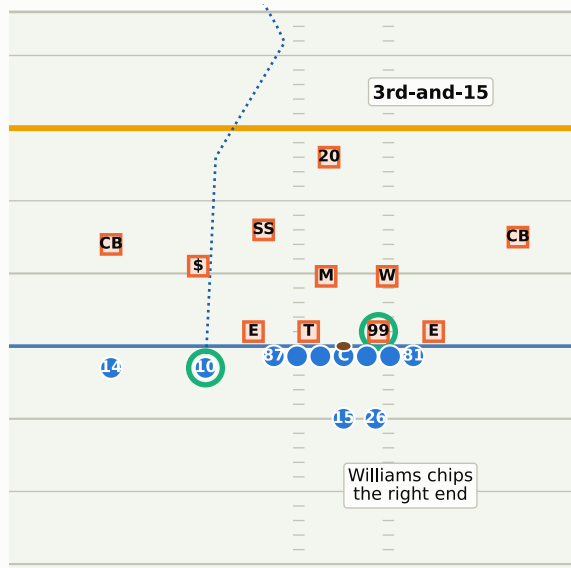
The scramble drill turns a broken play into a second play with its own rules, and the second play favours the offense. Man defenders run with their backs to the ball, so when their receivers change direction they lose a step turning to find them. Zone defenders have dropped to spots chosen for a throw that was supposed to come at two and a half seconds, and the spots are now wrong. The cost is on the offense's side too: receivers have to learn the rules and trust that the quarterback follows them, and a quarterback who extends plays will take sacks and make some awful throws. For most passers the trade is a loss. For a few it is a weapon, and Kansas City built around one.

Signature play: Jet Chip Wasp

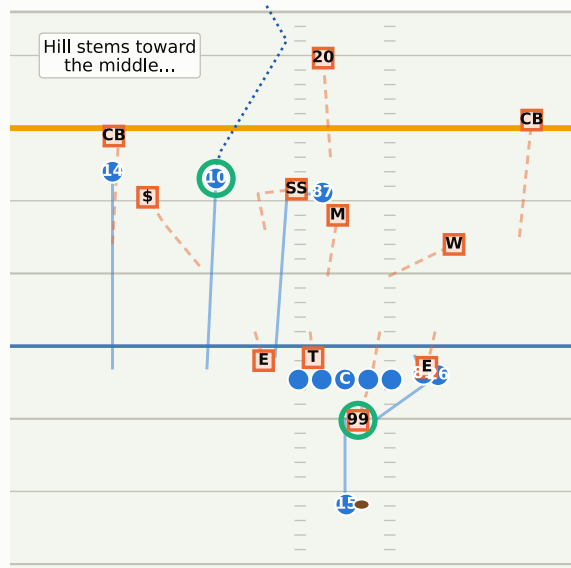
Now the hook makes sense. Third-and-15 is the down where defenses are most comfortable: they can rush four, drop seven, and keep everything in front of them, because 14 yards of gain is still a win for them. San Francisco did exactly that: four rushers, Cover 3 behind them, with two linebackers and a nickel inside the 49ers' five defensive backs.¹¹ Here is the play, as reconstructed from the charting and the published accounts.

¹¹nflverse participation (NGS charting) and play-by-play, game [2019_21_SF_KC](#), fourth quarter, 7:13, play 3086: shotgun; offensive personnel one back, two tight ends, two receivers; defensive personnel four linemen, two linebackers, five defensive backs; four pass rushers; coverage COVER_3 (zone); time to throw 3.778 seconds; pressure charted; targeted route GO; 43 air yards; "P.Mahomes pass deep left to T.Hill to SF 21 for 44 yards (J.Ward) [D.Buckner]" (the brackets credit Buckner with the quarterback hit). The 22 players on the field from the participation file's player IDs matched to nflverse rosters (Kansas City: Mahomes 15, D. Williams 26, Kelce 87, Bell 81, Hill 10, Watkins 14; San Francisco's defensive line Ford, Bosa, Armstead and Buckner 99, linebackers Warner and Greenlaw, defensive backs Sherman, Moseley, K. Williams, Tartt and Ward 20).

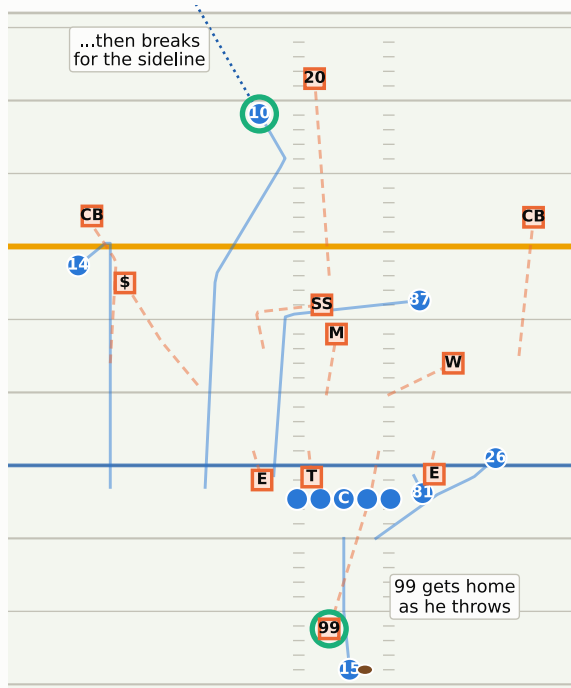
Snap · trips left, Kelce in-line



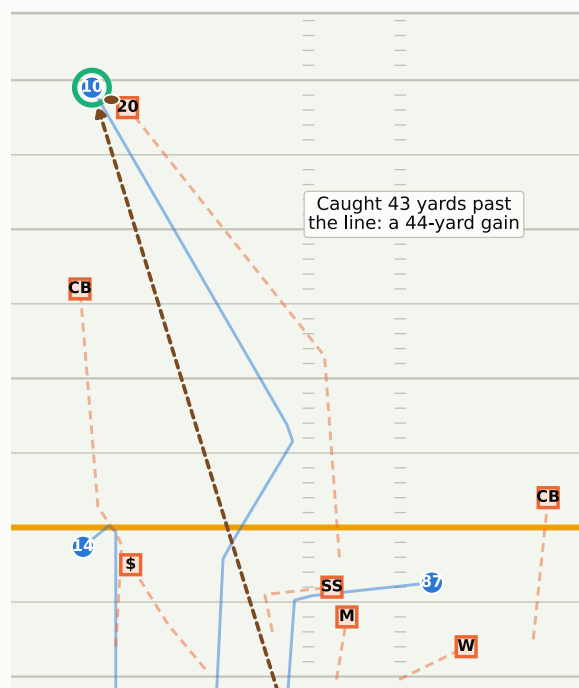
+1.8 s · the crosser stem



+3.8 s · the throw, 14 yards deep



+7.0 s · the stinger



- Offense (Chiefs)
- ◇ Defense
- Path so far (offense)
- - - Path so far (defense)
- - - Pass

Figure 5: ‘2-3 Jet Chip Wasp,’ Super Bowl LIV, third-and-15 at the Kansas City 35, 7:13 left, Chiefs trailing 20–10 (the panels follow the ball, so the second row shows more of the field than the first). Charted facts: shotgun with two tight ends (Kelce, 87, and Blake Bell, 81), Hill (10) and Sammy Watkins (14) to the same side as Kelce; four 49ers rushers; Cover 3; the ball out 3.8 seconds after the snap with Mahomes under pressure, DeForest Buckner (99) hitting him as he threw; caught 43 yards downfield and tackled by Jimmie Ward (20) for a 44-yard gain. Hill’s route is drawn from the published descriptions: a deep crosser’s stem toward the middle, then a sharp break back toward the sideline (the dotted line shows where he is going). The other routes, the protection and the defenders’ exact paths are our illustration, chosen to match those facts, not traced from film.

Take the call one word at a time, because each word is a piece of the answer to third-and-15.

“Chip.” A route this long needs time, so the protection is built to buy it. With six men blocking (five linemen and the second tight end) and the back hitting the edge rusher before he releases, the 49ers’ four rushers face seven blockers’ worth of attention for the first second and a half. Even so, Buckner beat his man, which tells you how close the margin is.

“Wasp.” Hill’s route starts like a deep crosser, angling toward the middle of the field at 15 to 20 yards, which is exactly where a Cover 3 defense wants a receiver to go: into the zone of the deep-middle safety, under the eyes of the linebackers. Ward, the deep-middle safety, does his job and takes the crosser. Then Hill breaks the other way, flattening toward the sideline at 20-plus yards, and the defense’s structure works against it. The deep-middle safety is now on the wrong side of him, and the left corner, whose deep third Hill is entering, has been keeping an eye on the receiver outside him (in our illustration, Watkins on a deep comeback). The ball is aimed for grass behind the corner and in front of the sideline.

The drift. Mahomes dropped and then kept retreating, to about 14 yards behind the line. That’s why the throw travelled about 57 yards in the air for a catch 43 yards downfield. Most quarterbacks can’t make that throw from that spot with a defensive tackle closing, and most offenses wouldn’t call a route that needs it. This is the defensive arithmetic of the previous section in one snap: the 49ers’ coverage held for three seconds, which is all a coverage is built to do, and the play wasn’t done yet.

i Madden vs. real life

In Madden you pick the play. On a real sideline the play-caller picks it, but a great quarterback’s opinion carries weight, and Reid’s staff lets Mahomes ask for plays he believes in. “Is there time for Wasp?” is a question about protection and about the clock, and the coaches had to agree before the call went in. The play existed because the staff had built it and the team had practiced it; the timing came from the player who would have to make it work.

Two more things to notice. First, Wasp is a **third-and-long** play, and having one is a statement. Most offenses treat third-and-15 as a down to survive: a screen, a draw, a checkdown, and punt. A West Coast team with an Air Raid quarterback treated it as a down to win. Second, the 49ers did almost nothing wrong. Four-man rush, Cover 3, deep safety on the crosser, pressure arriving. The defense lost to a design that asked for one more second than a defense is built to give and to a quarterback who could use it.

Era one: Tyreek Hill and the deep shot (2018–2021)

Mahomes’s first season as the starter was one of the best passing seasons in history: 5,097 yards, 50 touchdowns and the MVP award.¹² Kansas City finished first in the league in EPA per play. The shape of that offense is easy to describe: a West Coast menu underneath, Travis Kelce in the middle of the field, and Tyreek Hill, perhaps the fastest receiver in the league, threatening every deep third on every snap. In 2018 the Chiefs threw 15% of their passes 20 or more yards downfield, one of the highest shares in the league, and Hill was the target on 38% of the Chiefs’ deep throws from 2018 to 2021.¹³

¹²Mahomes’s 2018 (5,097 yards, 50 touchdowns, MVP) and 2022 (5,250 yards, 41 touchdowns, MVP) seasons and his three Super Bowl MVP awards (LIV, LVII, LVIII): “Patrick Mahomes,” Wikipedia, cited in the note on the draft; the 2022 totals agree with nflverse play-by-play.

¹³Course calculation from nflverse play-by-play, regular-season pass attempts (sacks and spikes excluded): Kansas City threw 15.1% of attempts 20 or more air yards in 2018 against a league median of about 12%; 8.9% in 2022; 13.3% in 2025. Deep targets 2018–2021: 308, of which Hill 118 (38%), Kelce 49 (16%), Demarcus Robinson 36 and Mecole Hardman 35; 2022–2024: 168, led by Justin Watson 41 (24%), Marquez Valdes-Scantling 32 (19%), Kelce 26 (15%) and Xavier Worthy 17 (10%). Hill’s trade to Miami (March 23, 2022, for five draft picks): NFL.com, “Chiefs trading WR Tyreek Hill to Dolphins for multiple draft picks, including 2022 first-rounder”: <https://www.nfl.com/news/chiefs-trading-wr-tyreek-hill-to-dolphins-for-multiple-draft-picks>. The 2018 AFC Championship Game (New England 37, Kansas City 31, overtime) from nflverse schedules.

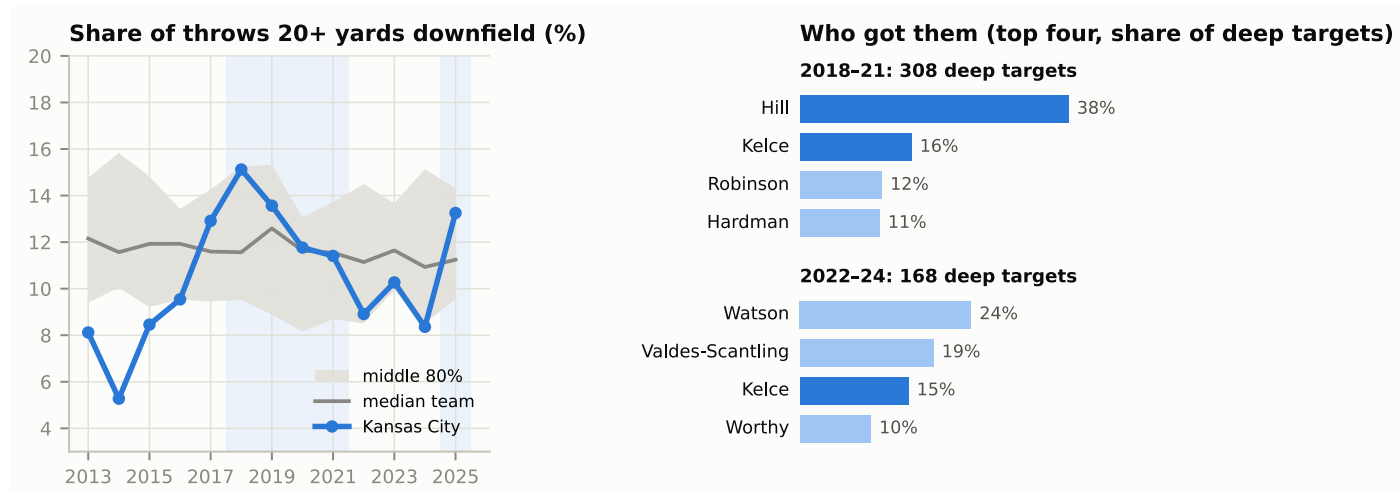


Figure 6: Two eras in one chart. Left: the share of Kansas City’s pass attempts thrown 20 or more yards past the line, against the league (grey band: middle 80% of teams; grey line: median team). The shaded columns mark the Mahomes-and-Hill seasons (2018–2021) and 2025. The deep share peaked in 2018 and fell as defenses adjusted; after Hill was traded in March 2022 it dropped below the league median, and it came back in 2025. Right: who got the deep targets. In the Hill era one receiver had 38% of them; in 2022–2024 they were spread across four receivers, none with more than a quarter. Regular seasons; sacks and spikes excluded.

Data: nflverse play-by-play.

Defenses answered the deep ball the way you’d expect: by putting a second safety deep. You met the numbers in *The Quick Game*: Kansas City saw a two-high shell on about 36% of its dropbacks in 2018 and about 46% in 2021, third-most in the league, and its average throw got shorter every year. Two-high coverage is a deal: it gives up room underneath and a lighter box against the run in exchange for never being beaten over the top. Mahomes’s 2019–2021 seasons are the story of a great quarterback learning to take that deal: the checkdown to the back, the 8-yard throw to Kelce, the run against six in the box. Deep shots still came (the Chiefs’ share stayed near the league median), but now they had to be earned with play-action, motion or a play like Wasp that manufactured a deep window against a defense built to deny one.

It worked well enough to reach three straight AFC Championship Games and two Super Bowls in that era. The first title game, after the 2018 season, was lost 37–31 in overtime to New England, the Patriots scoring on the first possession of overtime against a defense that had ranked 23rd of 32 in EPA per play allowed. Mahomes never touched the ball in overtime. Two days later Reid fired his defensive coordinator, and the other half of this story begins.

Era two: Kelce, the option route and the quick game (2022–2024)

In March 2022 the Chiefs traded Hill to Miami for five draft picks rather than pay him a receiver’s top-of-market contract. It was a cap decision as much as a football one (*Building a Roster* explains why quarterback-contract teams make these trades). The offense that came after it rebuilt itself around Travis Kelce and a rotating cast of receivers, and around a shorter passing game: the share of deep throws fell to 9% in 2022, and the average throw in 2023 and 2024 was the second-shortest in the league (*The Quick Game* has that chart). And in 2022, the first season without Hill, Mahomes threw for 5,250 yards and 41 touchdowns, won his second MVP, and the Chiefs finished first in EPA per play again. They won the Super Bowl.

The engine of that offense was the oldest idea in the passing game: a receiver who decides where to go after the snap. Kelce’s signature route, the option route, sends him to about 5 or 6 yards and lets him choose: against man coverage, break away from the defender’s leverage; against zone, find the window between two defenders and sit in it, facing the quarterback.

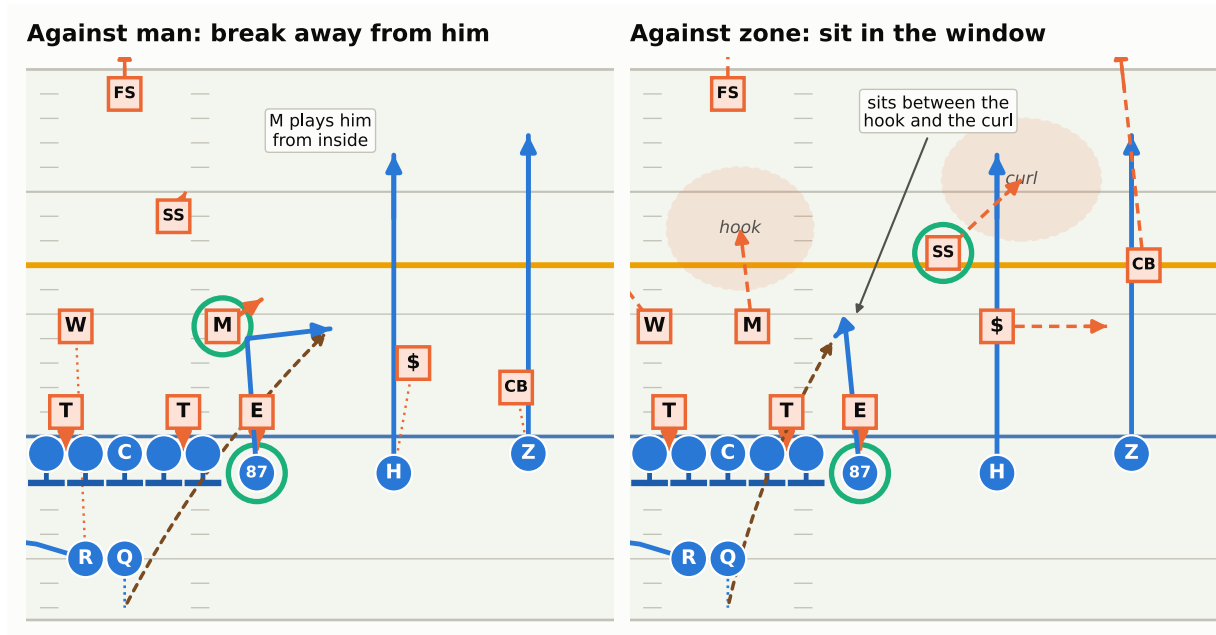


Figure 7: The option route, the heart of the Kelce era (illustrative): the tight end (87) is the innermost receiver of a trips set, and the two receivers outside him run vertical routes to clear space. Left, against man coverage: the Mike linebacker (M) plays Kelce from inside, so Kelce pushes to about 6 yards and breaks out, away from him, and the ball comes as he turns. Right, against zone (here Cover 3): there is no man to beat, only space. The Mike drops to the hook zone and the strong safety (SS) to the curl, and Kelce stops in the window between them and turns to face the quarterback. Same play call, two different routes, decided after the snap by what the defense does.

Why build an offense around a route like that? Because it can't be called wrong. A defensive coordinator's whole craft is matching his call to the offense's call: zone against the man-beaters, man against the zone-beaters. An option route doesn't have a call to match. It reads the coverage and becomes whatever beats it, and a receiver who has run it a thousand times with the same quarterback reads it at the same instant the quarterback does. Kelce, a third-round pick in 2013, ran it at a level that made him the tight end of his era (you met him in *The Twenty-Two*).

The trade is what you give up: depth and time. An option route is a 5-to-8-yard play. It moves the chains; it rarely changes a game by itself, and it depends on two players agreeing on a decision made in a split second. When they disagree, the ball goes to a spot the receiver isn't, and the interception that follows looks like the quarterback's fault on the broadcast. The quick game also asks the rest of the offense for patience: 12-play drives, with a dozen chances for a holding penalty or a dropped pass to end them.

That cost showed in 2023. Kansas City's wide receivers led the league in dropped passes, the offense slipped to 10th in EPA per play, and for the first time in the Mahomes years the Chiefs won with defense: Kansas City allowed 294 points, second-fewest in the league, and Spagnuolo's unit ranked 6th in EPA per play allowed.¹⁴ They won the Super Bowl anyway, beating San Francisco 25–22 in overtime. In 2024 they went 15–2, with an offense ranked 9th, by winning close games: 11–0 in regular-season games decided by eight points or fewer.¹⁵ Close-game records like that don't usually repeat, because one bounce of the ball in any of those games turns a win into a loss (Expected Points and Success Rate explains why a team's point differential predicts next season better than its record). The Chiefs reached a third straight Super Bowl and lost it badly.

The other half: Spagnuolo's pressure defense

The defense that lost the 2018 AFC Championship Game had ranked 31st in yards allowed. Four days after that loss, the Chiefs hired Steve Spagnuolo, a man he knew well: Spagnuolo had coached defensive backs and linebackers on Reid's Philadelphia staff from 1999 to 2006, under the great blitz coach Jim Johnson, before becoming the coordinator whose four-man rush beat the unbeaten

¹⁴Course calculation from nflverse schedules, 2023 regular season: points allowed, Baltimore 280, Kansas City 294, San Francisco 298. Offensive and defensive EPA ranks: see the EPA chart. The 2023 drops: "In 2023, their receiver position group led the NFL in drops (28) and drop rate (12%), according to SportRadar": Garrett Podell, CBS Sports, "Chiefs signing Marquise Brown: How ex-Cardinals speedster fits into Patrick Mahomes' offense in Kansas City" (Mar. 14, 2024): <https://www.cbssports.com/nfl/news/chiefs-signing-marquise-brown-how-ex-cardinals-speedster-fits-into-patrick-mahomes-offense-in-kansas-city>. Other charters' counts differ; all of them have Kansas City first.

¹⁵Course calculation from nflverse schedules, 2024 regular season: Kansas City 11–0 in games decided by eight points or fewer (6–5 in 2023).

Patriots in Super Bowl XLII. He had been out of football for the 2018 season.¹⁶ (Blitzing tells the Johnson and Giants story.)

In Kansas City, Spagnuolo became the most famous Cover 0 caller in the league. The famous thing is not how *often* he blitzes. It's what he blitzes *with*.

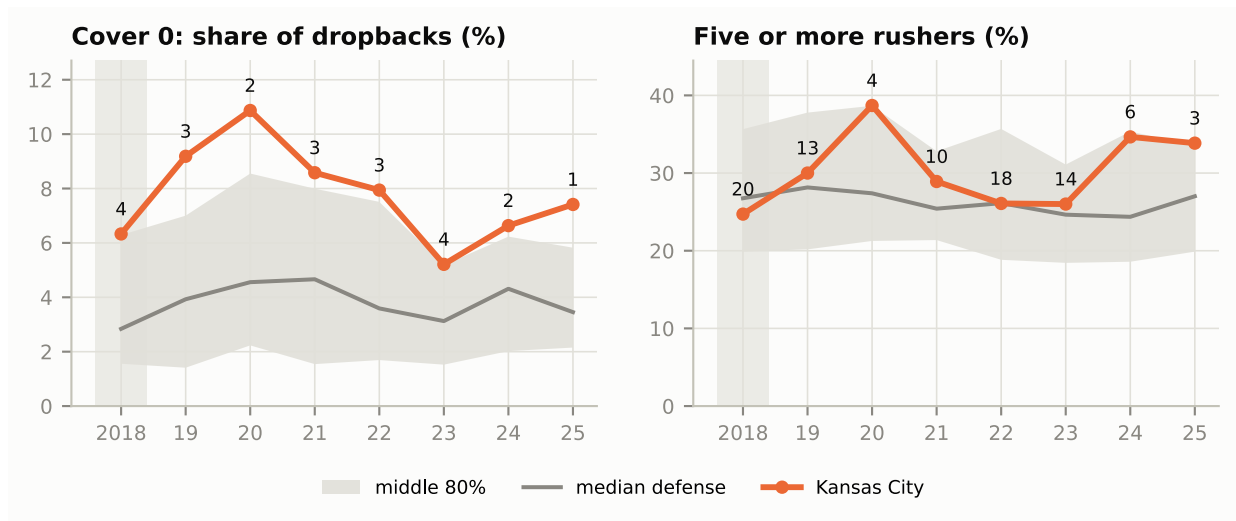


Figure 8: Spagnuolo's signature in two lines, 2018–2025 regular seasons (2018, shaded, was Bob Sutton's last season as coordinator). Left: the share of dropbacks on which the Chiefs played Cover 0, man coverage with nobody deep; right: the share on which they sent five or more rushers. Numbers by the dots are Kansas City's rank of 32. Under Spagnuolo the Chiefs ranked in the top four in Cover 0 every season, first in 2025, while their overall blitz rate was ordinary in most seasons (13th, 10th, 18th and 14th in 2019 and 2021–2023). Coverage labels are charted by NGS through 2022 and by FTN from 2023, so compare the seasons on either side of that line with care. Data: nflverse participation via nflreadpy.

From 2019 to 2025, Kansas City played Cover 0 on about 8% of charted dropbacks, roughly twice the rate of the rest of the league (4%), and the pattern in *when* is the interesting part. On fourth down the Chiefs played zero 27% of the time (everyone else, 14%); on third down 10% (6%); in the red zone 24% (18%). And outside the red zone, where most coaches almost never risk it because one missed tackle is a touchdown, they played zero nearly three times as often as everyone else (5.4% against 1.9%).¹⁷ Spagnuolo's zero isn't a habit you can game-plan by down and distance. It is a possibility on every snap, and that possibility is the weapon: an offense that has to protect against six or seven rushers on any down keeps a back or tight end in to block, which takes a receiver out of the pattern even on the snaps when the zero never comes.

¹⁶Charlotte Carroll, *Sports Illustrated*, "Report: Steve Spagnuolo Hired as New Chiefs Defensive Coordinator" (Jan. 24, 2019, four days after the AFC Championship Game; the team's announcement followed on Jan. 25): <https://www.si.com/nfl/2019/01/24/steve-spagnuolo-new-chiefs-defensive-coordinator>; CBS Sports, "Ex-Giants, Rams coach Steve Spagnuolo takes over as Chiefs defensive coordinator": <https://www.cbssports.com/nfl/news/ex-giants-rams-coach-steve-spagnuolo-takes-over-as-chiefs-defensive-coordinator> (Sutton fired on Tuesday, January 22, two days after the 37–31 overtime loss in the AFC Championship Game; Spagnuolo on Reid's Eagles staff 1999–2006; out of football in 2018; the 2018 Chiefs 31st in yards allowed per game).

¹⁷Course calculation from nflverse `load_participation()` coverage labels joined to play-by-play, 2019–2025 regular-season dropbacks with a charted coverage (BLOWN and COMBO excluded): Kansas City Cover 0 on 7.9% (331 dropbacks) against 4.2% for the league; by down, 5.2% / 7.9% / 9.6% / 27.4% on first through fourth down (rest of the league 2.7% / 3.6% / 5.9% / 13.9%); inside the opponent's 20, 24.0% (18.2%); outside it, 5.4% (1.9%). EPA per dropback against Kansas City's Cover 0: –0.048 (rest of the league's Cover 0: +0.071). Season ranks are in the chart. Labels come from NGS through 2022 and FTN from 2023 (FACTS caveat: the label sets differ across that break).

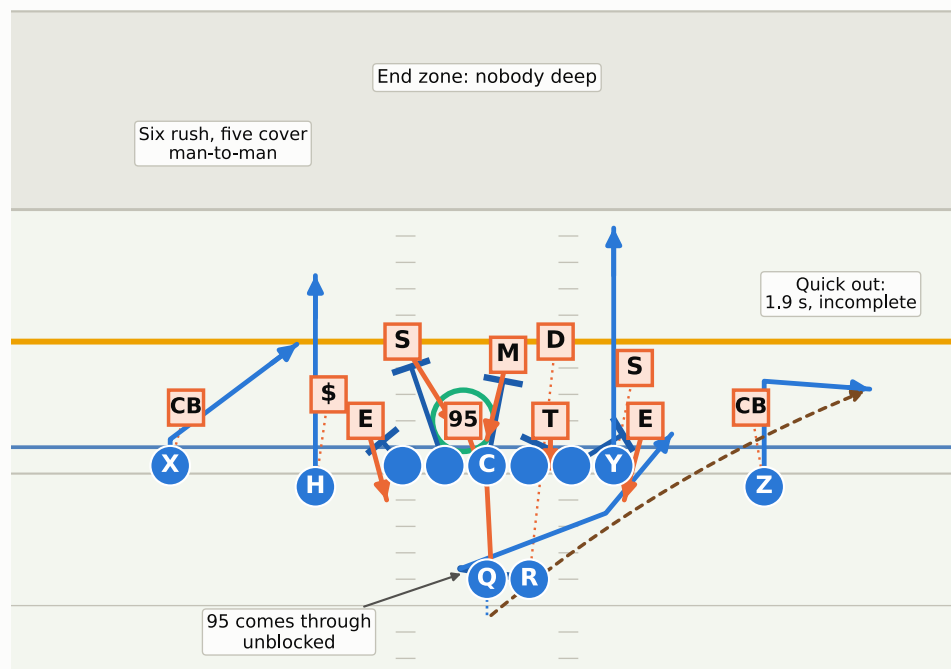


Figure 9: Spagnuolo’s goal-line zero, modelled on the third-and-4 at the Kansas City 9 in overtime of Super Bowl LVIII. Charted facts: 49ers in the shotgun with one back, one tight end and three receivers; a play-action fake; six Kansas City rushers, two of them blitzers; Cover 0 behind them; the pass out in 1.9 seconds, under pressure, short to the right, incomplete, with Chris Jones (95) credited with the hit. The exact blocking, the routes other than the throw and the defenders’ alignments are illustrative. With six rushing and five covering, every receiver has a man on him (dotted lines) and nobody is deep: the end zone behind them is empty. The protection has six blockers for six rushers on paper, but the running back is carrying out his fake when the rush arrives, and one rusher comes through clean.

That snap is the Cover 0 bargain at its best for the defense.¹⁸ Near the goal line the field is short, so the deep ball that punishes zero everywhere else barely exists; the throw has to come out in under two seconds; and one free rusher is enough. The 49ers kicked a field goal, Mahomes drove for the winning touchdown, and the 49ers never got the ball back. Notice also how Kansas City’s zeros fared over the whole period: from 2019 to 2025 opponents averaged about -0.05 EPA per dropback against them, while everyone else’s Cover 0 gave up about $+0.07$. That’s a sample of only 331 Kansas City zeros, so treat the gap as a hint, not a law, but it fits the idea that *when* you call zero matters as much as how often.

! Why it exists

Why does a Mahomes offense want a pressure defense next to it? Because of what a lead does to an opponent. A team trailing the Chiefs has to pass, and passing downs are where pressure pays most: the offense can’t keep extra blockers in without leaving receivers on the bench, and it can’t hand off to punish a blitz. A defense that can produce a sack or a hurried incomplection on a chosen snap converts the offense’s points into the opponent’s desperation. The price is variance. Zero gives up long touchdowns when one tackle is missed or one receiver wins, and Spagnuolo’s units, outside 2023, were middle-of-the-pack defenses over a full season (12th to 21st in EPA allowed). He traded steadiness for a handful of plays a game that end drives, and that suits a team whose offense will score anyway.

The data fingerprint

Here is the whole Reid era in one chart, offense and defense, with the eras marked.

¹⁸nflverse play-by-play, participation and FTN charting, game 2023_22_SF_KC, overtime, 7:29, third-and-4 at the Kansas City 9: “B.Purdy pass incomplete short right [C.Jones]”; shotgun, one back, one tight end, three receivers; six pass rushers (FTN: two blitzers); coverage COVER_0 (man); play-action and motion charted; time to throw 1.9 seconds; pressure charted. Jones’s untouched rush and the players’ quotes: Barry Petchesky, “The Chiefs Defense Unleashed Hell,” Defector (Feb. 12, 2024): <https://defector.com/the-chiefs-defense-unleashed-hell> . Kansas City’s Cover 0 EPA comparison: see the note on Cover 0 rates.

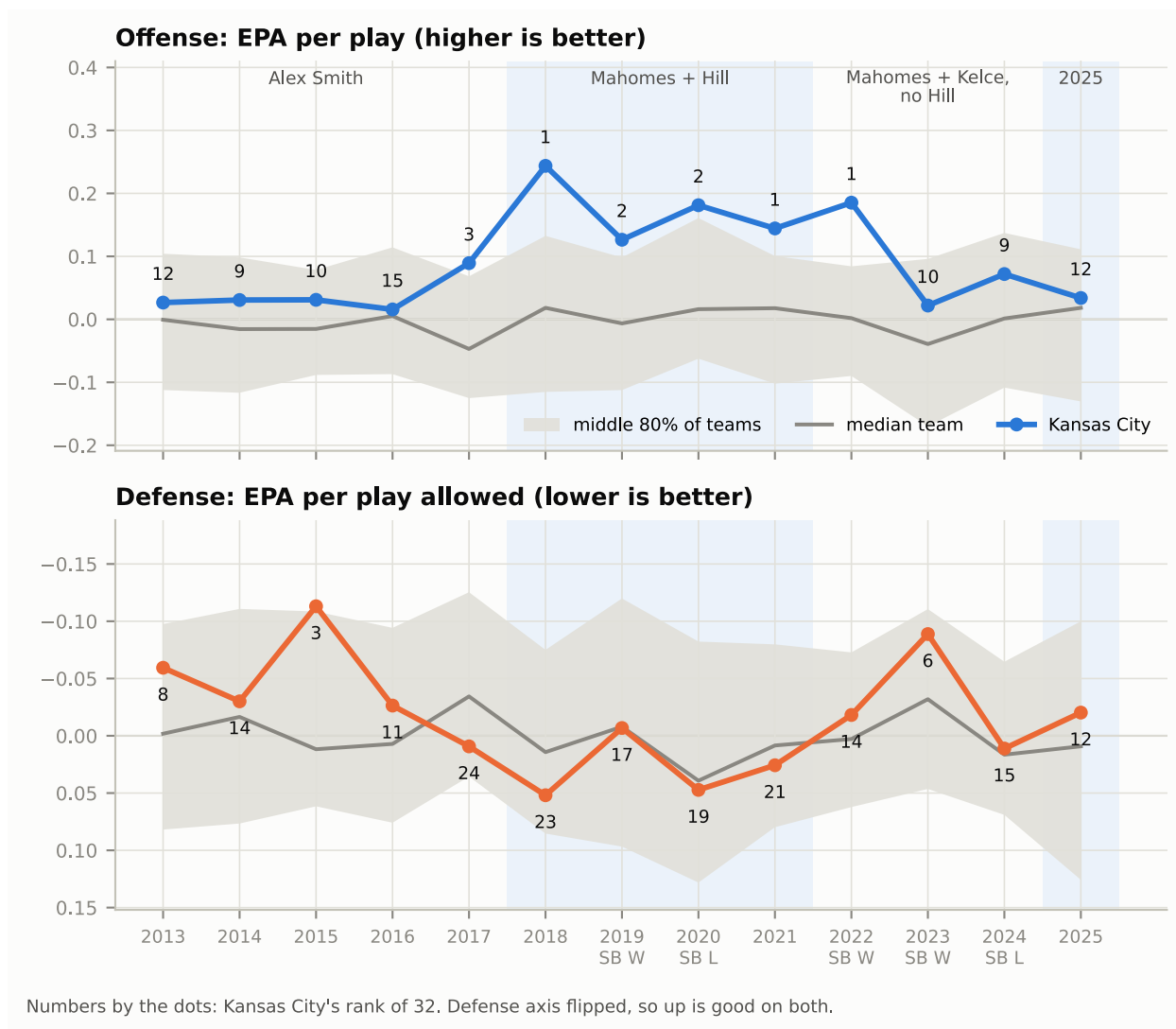


Figure 10: Kansas City's EPA per play by season, 2013–2025 regular seasons, against the league (grey band: middle 80% of teams; grey line: median team). Numbers by the dots are the Chiefs' rank of 32. Top, offense: good but not elite with Alex Smith (9th to 15th from 2013 to 2016), third in 2017 when the RPO-heavy offense took off, first or second every season from 2018 to 2022 with Mahomes, then 10th, 9th and 12th from 2023 to 2025. Bottom, defense (axis flipped so that up is good in both panels): mostly in the bottom half until 2022, sixth in 2023, the season the defense carried the team. SB W and SB L mark the seasons that ended in a Super Bowl win or loss. Data: nflverse play-by-play (plays with an EPA value; pass and run plays only).

Two things stand out. The offense's peak, five straight seasons of first or second from 2018 to 2022, spans both receiving eras: the Hill trade changed the *shape* of the offense, not its level, which is the strongest evidence that the system and the quarterback, not one receiver, were the engine. And the three Super Bowl wins came with three different balances: an elite offense and a middling defense (2019), an elite offense and an average defense (2022), and a good defense carrying a middling offense (2023).

Beyond EPA, a team's fingerprint is the set of things it does more or less often than everyone else. The table pools the seasons for which each measure exists.

Table 1: The Chiefs’ fingerprint. Rank of 32, 1 = most often. Seasons pooled, regular season only. Shotgun and deep-throw rates from nflverse play-by-play; RPO, screen, play-action, motion, trick-play and out-of-pocket flags from FTN charting (motion means motion before or at the snap); Cover 0 and rusher counts from nflverse participation (NGS through 2022, FTN from 2023).

Measure (share of plays)	Seasons	Chiefs	Median team	Chiefs’ rank
Shotgun snaps	2018–25	80.3%	67.5%	3
RPOs	2022–25	9.5%	4.6%	1
Screens	2022–25	7.7%	5.2%	1
Play-action	2022–25	11.4%	13.2%	27
Motion (before or at snap)	2022–25	53.9%	46.6%	6
Trick plays	2022–25	0.8%	0.7%	16
Left the pocket (dropbacks)	2022–25	22.7%	17.0%	6
Deep throws (20+ yd)	2018–21	12.9%	11.6%	7
Deep throws (20+ yd)	2022–24	9.2%	11.4%	29
Cover 0 (defense)	2019–25	7.9%	3.9%	2
5+ rushers (defense)	2019–25	31.0%	27.0%	5

Read the table as a portrait. Shotgun, RPOs and screens: the college graft and the West Coast core, done more than almost anyone. Play-action: *less* than most teams, the opposite of the Shanahan offenses, which is a reminder that “West Coast family” describes a language, not a set of plays (Offensive Systems makes the same point from the other side). Motion: heavy through 2024, though by 2025 the league had caught up. Trick plays: rare, as the misconception box said. And on defense, a team that blitzes somewhat more than average (about 31% of dropbacks against 27%, driven by three seasons) but plays Cover 0 about twice as often as the median defense: the zero, not the blitz count, is the signature.

How opponents responded

A team that wins for seven years teaches the league how to beat it, slowly and expensively. Four answers stand out, and each is a lesson about the arithmetic from earlier in the chapter.

1. Two deep safeties (2019–2021). The first answer was to take away Hill: two-high shells, and the patience to let Kansas City drive the field in small pieces. It didn’t stop the Chiefs; it changed them. The deep share fell, the throws got shorter, and the offense kept ranking first or second.

2. Rush three, drop eight (January 2022). Cincinnati’s defensive coordinator Lou Anarumo found the counter-intuitive answer to a quarterback who beats the clock: stop trying to beat the clock.

¹⁹Course calculation from nflverse play-by-play and participation, game 2021_21_CIN_KC (Jan. 30, 2022): Kansas City led 21–10 at halftime and lost 27–24 in overtime; three or fewer pass rushers on 5 of 22 charted Mahomes dropbacks in the first half and 10 of 24 in the second half and overtime; EPA per dropback +0.60 and –0.78. The 21–3 second-quarter lead, Mahomes’s 18-of-21 first half (220 yards, three touchdowns), Vonn Bell’s overtime interception and Evan McPherson’s winning field goal: Associated Press, “Mistakes on offense cost Chiefs 3rd straight Super Bowl trip” (Jan. 30, 2022), via WJXT: <https://www.news4jax.com/sports/2022/01/30/mistakes-on-offense-cost-chiefs-3rd-straight-super-bowl-trip/>. The eight-man coverage, the brackets on both Kelce and Hill and the spy: Mike Santagata, *Sports Illustrated* (All Bengals), “Dropping 8: How Lou Anarumo Stifled Patrick Mahomes and the Chiefs’ Offense” (Jan. 31, 2022): <https://www.si.com/nfl/bengals/allbengals-insiders-plus/how-bengals-shut-down-patrick-mahomes-chiefs-offense>.

The game. January 30, 2022, at Arrowhead. Kansas City led 21–3 in the second quarter and 21–10 at halftime; Mahomes had completed 18 of 21 passes for three touchdowns. In the second half and overtime Cincinnati rushed three or fewer far more often (on 10 of Mahomes’s 24 charted dropbacks, against 5 of 22 before halftime), dropped eight into coverage, bracketed Kelce and Hill, and kept a defender on Mahomes. The Chiefs’ offense went from about +0.60 EPA per dropback in the first half to about –0.78 after it, scored three points, and lost 27–24 in overtime after Mahomes threw an interception on the first possession.¹⁹

What to notice: on the three-man rushes, the pocket stays clean and Mahomes has all the time in the world, and it doesn’t help him. Count the defenders in coverage and find the extra one. With eight dropping, every receiver who comes open does so in front of a second defender, and the scramble drill has nowhere to go because a spy is waiting for the scramble. The rush can’t win in three seconds, so the defense stops trying to and covers for six.

The weakness of the answer is the same as its strength: three rushers rarely get home, so it only works with disciplined coverage players and enough of them, and a quarterback who stays patient can still dink his way down the field. It’s a plan for one game or a few drives at a time.

3. Win up front with four (Super Bowls LV and LIX). The two Super Bowl losses look alike. In February 2021 Tampa Bay’s Todd Bowles played two-high coverage on 43 of Kansas City’s 49 charted dropbacks and rushed four on 50 of 56. The Chiefs were without both starting offensive tackles (Eric Fisher had torn his Achilles in the AFC Championship Game; Mitchell Schwartz had been out since the autumn with a back injury), and the Buccaneers pressured Mahomes on a Super Bowl–record 29 of his 56 dropbacks; by NFL Next Gen Stats’ count he ran 497 yards before throwing or being sacked. Tampa Bay won 31–9.²⁰ Four years later, in Super Bowl LIX, Vic Fangio’s Eagles did it again: four rushers on 40 of 45 charted dropbacks, quarters coverage on 27 of them, six sacks, and a 40–22 win (A First Look at Defense and The Pass Rush tell that game from the defense’s side).²¹

That’s the most complete answer anyone found, and the most expensive: you need a defensive front good enough to pressure Mahomes with four, so that seven can cover, two of them deep. Blitzing him didn’t work (he sees it and throws hot); covering with eight didn’t pressure him. Four rushers who win their one-on-ones, in disciplined lanes so he can’t escape, plus two-high coverage behind them, attacks both halves of the arithmetic at once. Few teams have that front. The ones that did beat him in the biggest games.

4. Time. The last answer is no team’s plan. Kelce turned 36 during the 2025 season, the receivers around him changed every year, and the offensive line was rebuilt more than once. In 2025 the Chiefs ranked 12th in EPA per play, 6–11 overall, and on December 14, in a 16–13 home loss to the Chargers, Mahomes tore the ACL in his knee late in the game. He had surgery the next day. The loss eliminated Kansas City from playoff contention for the first time since 2014, and it was the team’s first losing season since 2012.²²

²⁰Course calculation from nflverse play-by-play and participation (NGS charting), game 2020_21_KC_TB: 56 Kansas City dropbacks, 50 against a four-man rush; of 49 with a coverage label, 43 two-high (quarters 21, Cover 2 nine, Cover 6 seven, 2-Man six) and six Cover 3. Fisher’s torn Achilles: CBS Sports, “Chiefs’ Eric Fisher will miss 2021 Super Bowl after tearing Achilles in AFC Championship Game”: <https://www.cbssports.com/nfl/news/chiefs-eric-fisher-will-miss-2021-super-bowl-after-tearing-achilles-in-afc-championship-game-per-report>; Schwartz’s back injury: NFL.com, “Chiefs OT Mitchell Schwartz undergoes back surgery”: <https://www.nfl.com/news/roundup-chiefs-ot-mitchell-schwartz-undergoes-back-surgery>. Pressured on 29 of 56 dropbacks, a Super Bowl record: “Patrick Mahomes,” Wikipedia (cited in the note on the draft). The 497 yards (“Mahomes traveled 497 yards before throwing the ball or being sacked ... the most scramble yards in a game by any quarterback in the Next Gen era (since 2016)”: Nick Shook, NFL.com, “Chiefs’ repeat bid doomed by struggles up front in Super Bowl LV loss to Buccaneers” (Feb. 2021): <https://www.nfl.com/news/chiefs-repeat-bid-doomed-by-struggles-up-front-in-super-bowl-lv-loss-to-buccaneer>.

²¹Course calculation from nflverse play-by-play and participation (FTN charting), game 2024_22_KC_PHI: 45 Kansas City dropbacks, 40 against four rushers, four against three and one against five; coverage charted as quarters (COVER_4) on 27; six sacks. NBC News reported that the Eagles “blitzed Mahomes a grand total of zero times on his 42 dropbacks” (Rohan Nadkarni, Feb. 10, 2025: <https://www.nbcnews.com/sports/nfl/eagles-made-patrick-mahomes-look-mortal-rcna191414>); the one five-man rush in the charting is a different counting rule. Result and halftime score: Super Bowl LIX, this book’s fact sheet (Eagles 40, Chiefs 22) and “Super Bowl LIX,” Wikipedia.

²²“2025 Kansas City Chiefs season,” Wikipedia: https://en.wikipedia.org/wiki/2025_Kansas_City_Chiefs_season (6–11, third in the AFC West; Mahomes’s season-ending ACL tear late in the Week 15 game against the Chargers on December 14, a 16–13 home loss; eliminated from playoff contention for the first time since 2014; first losing season since 2012). Surgery the next day, to both the ACL and the LCL: “Patrick Mahomes,” Wikipedia. nflverse play-by-play shows his last 2025 snap in that game. Kelce’s age: born October 5, 1989 (“Travis Kelce,” Wikipedia). 2025 offensive EPA rank: see the EPA chart.

The Super Bowl ledger

Table 2: The five Super Bowls of the Mahomes era, three won and two lost. Kansas City reached seven straight AFC Championship Games (after the 2018 to 2024 seasons) and hosted six of them.²³

Game son)	(sea-	Date and site	Result	MVP	What decided it
LIV (2019)		Feb. 2, 2020, Hard Rock Stadium, Mi- ami Gardens	Chiefs 31, 49ers 20	Mahomes	Down 20–10 in the fourth quarter; Jet Chip Wasp on third-and-15 started three touchdown drives in the last seven minutes.
LV (2020)		Feb. 7, 2021, Raymond James Sta- dium, Tampa	Buccaneers 31, Chiefs 9	Tom Brady	Two backup tackles; record pressure from a four-man rush with two-high coverage behind it.
LVII (2022)		Feb. 12, 2023, State Farm Stadium, Glendale	Chiefs 38, Eagles 35	Mahomes	Mahomes on an injured ankle; two fourth-quarter “Corn Dog” touchdowns against man coverage.
LVIII (2023)		Feb. 11, 2024, Allegiant Sta- dium, Las Ve- gas	Chiefs 25, 49ers 22 (OT)	Mahomes	Spagnuolo’s zeros in the fourth quarter and overtime; the first Super Bowl under the 2022 overtime rule, won on a touchdown after the 49ers’ field goal.
LIX (2024)		Feb. 9, 2025, Caesars Su- perdome, New Orleans	Eagles 40, Chiefs 22	Jalen Hurts	Pressure from four rushers on almost every dropback; Kansas City trailed 24–0 at halftime.

What survives

The Chiefs went into 2026 with the same head coach, quarterback and defensive coordinator who started this story, and with an old coordinator back: Eric Bieniemy, the coordinator from 2018 to 2022 (he left for Washington in 2023), returned in January 2026 after Matt Nagy left to become the Giants’ offensive coordinator.²⁴ Whether Mahomes returns at his old level from a torn ACL, and whether a Reid offense can find its next Kelce, are 2026 questions this book can’t answer. What it can say is which parts of the Chiefs’ approach have already spread.

- **The marriage of the West Coast and the spread.** In 2017 the RPO was a college curiosity in the NFL. By the mid-2020s every offense carried some, and the idea that a West Coast language and an Air Raid quarterback could share a huddle is no longer odd. Reid’s version showed that a coach doesn’t have to choose between his system and the college game; he can add one to the other.
- **The off-script quarterback as a plan.** Teams now draft for the late throw: arm talent on the move and the instinct to extend a play are scouted as skills, not tolerated as flaws, and offenses teach the scramble drill as a play of its own (Quarterback Play).
- **The motion and the menu.** Reid didn’t invent pre-snap motion, but his offenses made a huge, motion-heavy, weekly-renewed playbook look normal, and the league’s motion rate climbed every year from 2018 to 2025 (Motion and Shifts).²⁵
- **Zero as a weapon of timing.** Spagnuolo’s lesson for defenses is that pressure is about *when*, not how often: a blitz rate only modestly above average, with a credible zero on any snap.

²³Super Bowls LVIII and LIX: this book’s fact sheet (planning/FACTS-current.md, §1). LIV, LV and LVII: “Super Bowl LIV,” “Super Bowl LV” and “Super Bowl LVII,” Wikipedia (dates, sites, scores and MVPs; Brady MVP of LV; Mahomes playing LVII on an injured ankle). Seven straight AFC Championship Games and the six at home: course calculation from nflverse schedules (the 2023-season game was at Baltimore). LIX halftime score (Eagles 24, Chiefs 0, after Hurts’s touchdown run, Cooper DeJean’s 38-yard interception return and Hurts’s pass to A.J. Brown): “Super Bowl LIX,” Wikipedia; Reuters and Associated Press game reports.

²⁴This book’s fact sheet (planning/FACTS-current.md, §0 item C4 and §9): Matt Nagy, Chiefs offensive coordinator 2023–2025, is the Giants’ offensive coordinator in 2026; Eric Bieniemy returned as Chiefs offensive coordinator in January 2026; Steve Spagnuolo remains defensive coordinator. ProFootballTalk, “Chiefs finalize deal with Eric Bieniemy to return as their OC”: <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/chiefs-finalize-deal-with-eric-bieniemy-to-return-as-their-oc>. Bieniemy was Washington’s offensive coordinator in 2023 (“Eric Bieniemy,” Wikipedia); the chapter does not trace his 2024–2025 stops.

- **The answer, too.** The defensive blueprint that beat Kansas City in its two Super Bowl losses (four rushers, two deep safeties, no blitz) is the defense of the mid-2020s, and Case Study: Fangio's Two-High Revolution and Macdonald's Answer tells that story.

And there is the coaching tree (the coaches a head coach trained who went on to lead teams of their own). From Reid's Philadelphia and Kansas City staffs came Super Bowl-winning head coaches John Harbaugh (Baltimore) and Doug Pederson (Philadelphia), plus Sean McDermott, Ron Rivera, Matt Nagy and Brad Childress, all NFL head coaches.²⁶ Reid's own career is the most durable survivor of all: a coach who learned the West Coast offense from Walsh's students in 1992, and who was still changing it thirty years later.

Watch for it on Sunday

💡 Watch for it: the off-script counter

Pick one quarterback this weekend (any team; the Chiefs if you can) and, for one half, sort every dropback into one of two piles.

- **In structure:** the ball comes out from the pocket on time. A rough clock: you say "one-thousand-one, one-thousand-two, one-thousand-three" from the snap, and it's gone before you finish.
- **Off script:** the quarterback leaves the pocket (outside the tackles), *or* he's still holding the ball at "one-thousand-four," *or* the receivers have clearly abandoned their routes and are running the scramble drill.
- **For each play, mark the result:** success (half the yards needed on first down, 70% on second, all of it on third or fourth: the success rate rule from Downs, Distance, and Field Position) or not, plus every sack, throwaway and turnover.
- **Compare the piles.** For most quarterbacks the off-script pile is smaller and worse. A pile of off-script plays that is nearly as good as the in-structure pile is the Mahomes signature, and it tells you the defense's coverage clock is wrong for this opponent.
- **Then watch the defense adjust.** Does it start rushing three and spying? Rushing four in tight lanes? Blitzing to force the ball out? Write down which, and whether the off-script pile shrinks.

Add one line to your prediction log for each half: *in-structure n / successes, off-script n / successes, what the defense changed.*

²⁵League motion rate (pre-snap shifts and motion, share of plays) from PFF: 43.0% in 2018, 46.8% in 2019, 50.0% in 2020, 51.8% in 2021, 55.4% in 2022, 56.4% in 2023, 61.5% in 2024 and 63.9% in 2025: Daire Carragher, PFF, "NFL pre-snap motion usage reaches new highs across the NFL" (Feb. 23, 2026): <https://www.pff.com/news/nfl-pre-snap-motion-usage-reaches-new-highs-across-the-nfl> (this book's fact sheet, §10).

²⁶John Harbaugh coached special teams (1998–2006; Ray Rhodes hired him, Reid kept him) and then defensive backs (2007) in Philadelphia before Baltimore hired him in January 2008; he won Super Bowl XLVII. Doug Pederson was on Reid's Eagles staff (2009–2012) and his Kansas City offensive coordinator (2013–2015), and won Super Bowl LII as Philadelphia's head coach. Sean McDermott (Eagles 1999–2010, coordinator 2009–2010; Bills), Ron Rivera (Eagles linebackers coach 1999–2003; Panthers, Washington) and Brad Childress (Eagles 1999–2005, coordinator from 2002; Vikings) also worked on Reid's Eagles staffs; Matt Nagy (Bears) was on both, an Eagles assistant from 2008 and a Chiefs coach from 2013 to 2017. Sources: the coaches' Wikipedia biographies ("John Harbaugh," "Doug Pederson," "Sean McDermott," "Ron Rivera," "Brad Childress," "Matt Nagy"). 2026 jobs: Harbaugh is the Giants' head coach and McDermott was fired by Buffalo in January 2026 (this book's fact sheet, §9).

Predict the play

💡 Predict the play 1: Kelce's choice

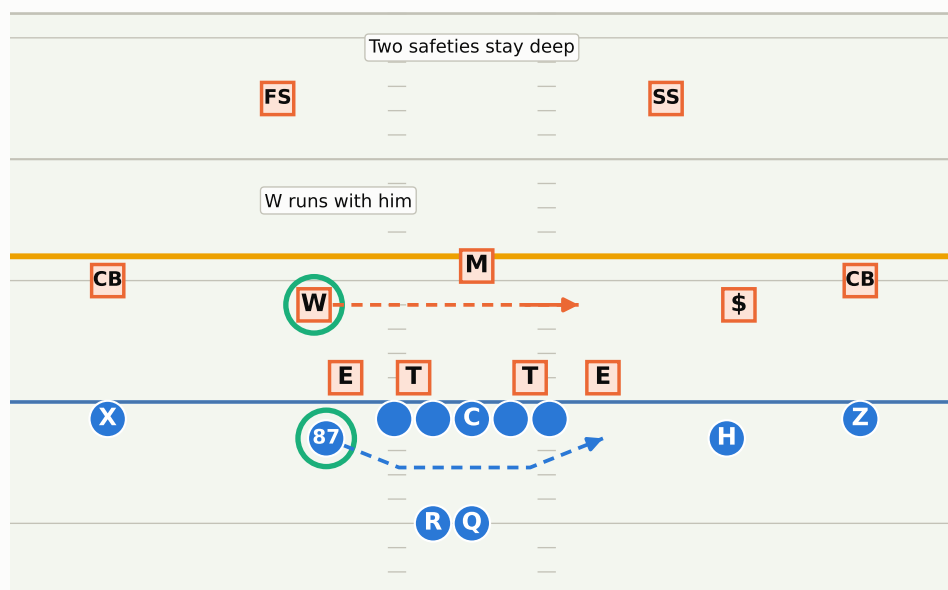


Figure 11: Predict the play 1. Third-and-6 near midfield. Kelce (87) starts as a wing on the left and motions across to become the inside receiver of a three-receiver set on the right (dashed blue arrow). The Will linebacker (W) runs across with him and settles just inside him (dashed orange arrow). Both safeties stay deep. Every arrow is movement before the snap.

Kelce is going to run his option route at about 6 yards, with the two receivers outside him running off. Before the snap, what has the defense told you about its coverage? Which way will Kelce break, and why?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: Spagnuolo on third down

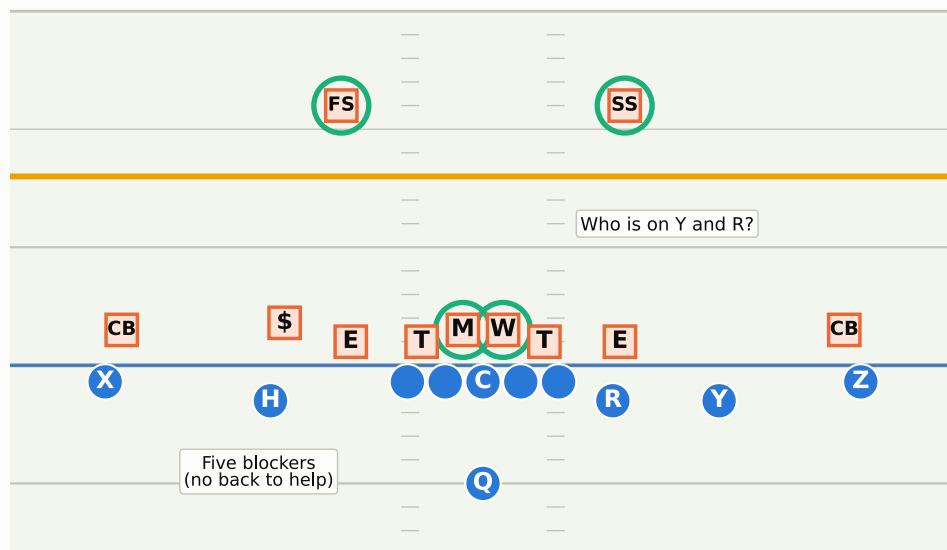


Figure 12: Predict the play 2. Third-and-8; the offense (not the Chiefs this time) is in an empty set, three receivers right, two left, with Kansas City on defense. Two linebackers stand in the A gaps either side of the center (M and W), four linemen beside them; the corners and the nickel (\$) press their receivers from a shade inside. Both safeties are about 11 yards deep. The two highlighted pairs are the questions: the linebackers in the gaps, and the safeties.

You're the quarterback. Count the rushers you might face, count your blockers, and work out who is covering your tight end (Y) and the back split out beside him (R). What is coming, and where does the ball have to go?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the shell that took away Hill

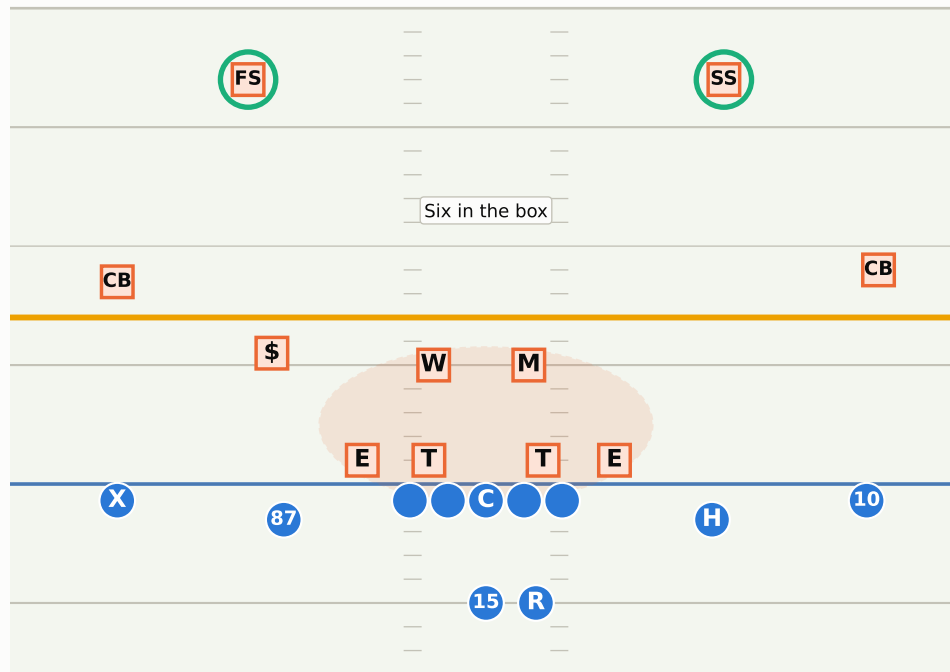


Figure 13: Predict the play 3. A 2021-style picture: second-and-7, Kansas City in the shotgun with three receivers and Kelce (87) in the slot on the left, Hill (10) wide right. The defense has two safeties about 17 yards deep and its corners 8 to 9 yards off the outside receivers. Count the defenders near the line (the box, shaded) and compare them with the Chiefs' blockers.

You're calling the plays for Kansas City. What do you call, and what are you willing to give up?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Reid built the Chiefs' offense from three sources:** West Coast bones (timed, horizontal passes, long calls, a weekly-renewed menu), the college spread and RPO game (studied deliberately from 2013, in full use by 2017, and an Air Raid quarterback in 2018), and a coach's appetite for plays built for one opponent. Shotgun, RPOs and screens were the fingerprint; trick plays were the seasoning.
- **Mahomes broke the defense's clock.** Defenses design coverage to hold for about three seconds; his throws after four seconds were worth as much as his on-time ones, while the league's are worth less than nothing. Every defensive answer (cover longer, rush in lanes, spy, blitz) costs something, and the scramble drill collects.
- **Two eras, one level.** The Hill years (2018–2021) were built on the deep threat, and defenses answered with two-high shells; the Kelce years (2022–2024) were built on the option route and the quick game. The offense ranked first or second in EPA per play in all five seasons from 2018 to 2022.
- **Spagnuolo's defense was the complement:** Cover 0 at roughly twice the league rate, ranked top four every season, with a blitz rate only modestly above average. Pressure as a matter of timing, not volume, and the defense that carried the 2023 title.
- **Opponents' best answer was four rushers and two deep safeties** (Tampa Bay in Super Bowl LV, Philadelphia in LIX), with Cincinnati's three-man rush as a one-game alternative. It needs a front that wins one-on-one, which is why few teams could use it.
- **The ledger:** Super Bowls LIV, LVII and LVIII won, LV and LIX lost, seven straight AFC title games; then 6–11 and a torn ACL in 2025, and Bieniemy back as coordinator for 2026.

Connections

Builds on: The Modern Era (the league the Chiefs helped make) and Blitzing (Cover 0 and the Jim Johnson tree), with Offensive Systems, Run-Pass Options, The Quick Game, Quarterback Play and Man Coverage underneath.

Sets up and connects to: the other case studies, especially Walsh’s 49ers (where Reid’s language came from), The Shanahan Tree (the other West Coast branch, with the opposite fingerprint), The Eagles (the team that beat the Chiefs in Super Bowl LIX) and Fangio’s Two-High Revolution (the defensive answer that spread). For measuring a team’s style yourself, see Tendencies, PROE and Charting; for why quarterback contracts force trades like Hill’s, Building a Roster.

Go deeper

- Chris B. Brown, *The Art of Smart Football* (2015): short essays on the West Coast offense, the spread and the packaged plays that became RPOs, the three sources of this chapter.
- Bill Walsh with Brian Billick and James Peterson, *Finding the Winning Edge* (1997): the system Reid learned from Holmgren, in the founder’s own words.
- Ted Nguyen’s film breakdowns of the Chiefs’ offense (The Athletic) and Seth Keysor’s Chiefs film work (Arrowhead Pride) for play-by-play detail on Reid’s designs.
- NFL Next Gen Stats (nextgenstats.nfl.com) for time to throw, pressure and scramble data season by season; this chapter’s own numbers come from nflverse and FTN through `nflready`, and the code is under “Show the Python.”

Answers to Predict the play

Answer 1.

Man coverage underneath, and Kelce breaks out, away from the Will. A linebacker who runs all the way across the formation with a motioning receiver is following a *man*, not guarding an area: the classic man indicator from Motion and Shifts. With both safeties staying deep, the likeliest call is 2-Man: man-to-man underneath, two deep halves over the top. The Will has settled *inside* Kelce, protecting against the in-cut and trusting the safety on his side to help over the top. So the option route’s rule for man applies: break away from the leverage. Kelce pushes to 6 yards, sells inside for a step, and breaks to the sideline, with Mahomes throwing as he turns, before the safety can drive down on it. If the Will had stayed home and the nickel had bumped inside to pick Kelce up, that would have said zone, and Kelce would have looked for the window to sit in.

Answer 2.

You can’t know, and that is the design; but you can know where the ball goes if it’s zero. Count it: four linemen and two linebackers in the gaps make six possible rushers against five blockers, and with an empty backfield there is no back to pick anyone up. If all six come, the five men left must cover five receivers, which means the two safeties 11 yards deep are the men on Y and R, and they will start sprinting down at the snap. That’s Cover 0 with one free rusher, and the ball has to be out before he arrives: to Y or R, who have 11 yards of space to work in for the first second, on a quick slant or a quick out.

But the same picture is also a simulated pressure waiting to happen: the two linebackers drop out of the gaps at the snap, two linemen drop or one crosses, four rush, and the hot throw you made in a hurry goes straight to a defender who was never blitzing (Zone Blitzes and Simulated Pressure). The base rates say zero is the minority call even for Spagnuolo: about 10% of the Chiefs’ third-down dropbacks from 2019 to 2025, more than half again the league’s rate but still one snap in ten. The pressed corners with inside leverage do tell you *man*, though, so a slant is risky and an out-breaking route has room. Set the protection to the mug, know your hot answer to Y or R, and if the safeties don’t come down, play the down normally.

Answer 3.

Take what the shell gives: a run, or a short throw, and save the shot. Two safeties at 17 yards and corners 9 yards off are saying one thing: nothing goes over our heads. The price the defense paid for that is in front of you. Only six defenders are in the box against five linemen and a running back, so an inside run has a blocker for nearly every defender, and the safeties, 17 yards away, arrive late. In the passing game the space is underneath: Kelce on an option route at 6 yards against a nickel with no help over him nearby, or a screen to the back. That’s what the Chiefs learned to do in 2019–2021, and what the drop in their deep-throw share in Figure 6 records.

What you give up is the explosive play, and with it some margin: a 10-play drive has more chances to stall than a 2-play one. The way to get the deep ball back is to make the safeties move first: play-action to pull them toward the line, motion to make one rotate, or a play like Wasp that gets behind a defense that thinks it is safe. Calling Hill’s go route into this shell because he is fast is the one thing the defense wants you to do.